

Eiken Syndrome: A Comprehensive Disease Characteristics Report

Disease: Eiken Syndrome (Eiken familial skeletal dysplasia) **MONDO ID:** MONDO:0010803 · **OMIM:** #600002 · **Orphanet:** ORPHA:79106 · **UMLS:** C1854486 **Category:** Mendelian (autosomal recessive skeletal dysplasia) **Causal gene:** *PTH1R* (HGNC:9608; OMIM *168468; NCBI Gene 5745; locus 3p21.31)

Summary

Eiken syndrome is an **ultra-rare autosomal-recessive skeletal dysplasia** caused by **biallelic (homozygous) hypomorphic/altered-function variants in *PTH1R***, the gene encoding the parathyroid hormone / parathyroid hormone-related peptide receptor type 1 (the PTH/PTHrP receptor). First described by Eiken and colleagues in 1984 in three brothers from a consanguineous Danish family, the disorder is defined radiographically by **severely delayed (retarded) endochondral ossification** — especially of the epiphyses, pelvis, hands and feet — together with **abnormal bone modeling, brachydactyly, coarse trabeculae, supernumerary/pseudo-epiphyses, and primary failure of tooth eruption**. Only about **four to five unrelated families** have been reported worldwide since 1984, spanning European, Indian, and East-African ancestries. Affected individuals have normal intelligence, normal lifespan, and are normal at birth, with skeletal abnormalities becoming apparent during childhood growth.

Mechanistically, Eiken syndrome occupies the **mild, non-lethal middle of the *PTH1R* allelic spectrum**. *PTH1R* is a class B G-protein-coupled receptor that, through the PTHrP–Indian hedgehog (Ihh) feedback loop, paces the proliferation and hypertrophic differentiation of growth-plate chondrocytes and therefore the timing of endochondral ossification. Eiken-associated mutations — distributed across the extracellular domain (E35K, Y134S), the transmembrane helices (I237N, D241E), and the intracellular C-terminal tail (R485X) — **bias receptor signaling**: cell-based assays show *increased basal cAMP tone* combined with *impaired β -arrestin2 recruitment and defective desensitization* of the PTHrP response. Because sustained PTHrP-type signaling delays chondrocyte hypertrophy (the mirror image of the accelerated ossification seen in complete loss-of-function Blomstrand lethal chondrodysplasia), the net

effect is **delayed ossification** — the phenotypic hallmark of Eiken syndrome. A transmembrane-helix-2 subset of variants additionally produces overt **PTH resistance** (hypocalcemia, hyperphosphatemia), placing those patients at the interface with pseudohypoparathyroidism-like disorders.

The evidence base is exclusively **case-report and in-vitro functional-assay level**; there are no registries, natural-history cohorts, quality-of-life instruments, or omics datasets specific to Eiken syndrome. Management is entirely **supportive** — calcium and active vitamin D for the PTH-resistant subset, orthopedic and dental care, and genetic counseling for consanguineous families. No targeted or curative therapy exists. Humanized *PTH1R* knock-in mice and conditional/global mouse models of the PTHrP-PTH1R axis reproduce key skeletal phenotypes and provide the primary in-vivo platform for the disease family.

Key Findings

F001 — Eiken syndrome is caused by biallelic hypomorphic *PTH1R* mutations, inherited autosomal-recessively

The original 1984 description of three affected brothers from a consanguineous family established autosomal recessive inheritance on clinical/pedigree grounds, with the authors noting that *"Parental consanguinity suggest an autosomal recessive inheritance"* (PMID: 6734674). Molecular confirmation came later: *"Eiken syndrome is a very rare skeletal dysplasia due to bi-allelic variants in PTH1R"* (PMID: 29987841). The reported causal alleles are homozygous and span multiple receptor domains — **R485X** (C-terminal tail truncation), **E35K** and **Y134S** (extracellular domain), and the newer transmembrane-helix-2 variants **I237N** and **D241E**. The second reported case carried homozygous **c.103G>A p.(Glu35Lys)** (PMID: 29987841). A recent review summarizing homozygous *PTH1R* disorders states that *"homozygous mutations located in the transmembrane helices, extracellular domains and C-tail of the PTH1R were identified in patients with milder conditions characterized by variable degrees of skeletal and mineral abnormalities. These include delayed ossification in Eiken syndrome"* (PMID: 40904804). The disorder is ultra-rare, with only ~4–5 families known.

F002 — Eiken mutations bias PTH1R signaling: increased basal cAMP with impaired β -arrestin recruitment

The defining molecular signature of Eiken syndrome is not simple loss or gain of function but a **rebalancing of the receptor's signaling outputs**. In HEK293 cell assays, *"R485X increases the receptor's basal rate of cAMP signaling and decreases its capacity to recruit β -arrestin2 upon ligand stimulation. The E35K and Y134S mutations each weaken the binding of PTHrP leading to impaired β -arrestin2 recruitment and desensitization of cAMP signaling response to PTHrP but not PTH"* (PMID: 37268817). The authors conclude that *"Our findings support a critical role for interaction with β -arrestin in the mechanism by which the PTH1R regulates bone formation"* (PMID: 37268817). The transmembrane-helix-2 variants behave similarly at the cAMP level: functional analysis *"demonstrated increased basal cAMP signaling for both variants, with relative blunting of responses to both PTH and PTH-related peptide (PTHrP) ligands"* (PMID: 39276366). The common thread is **elevated ligand-independent (basal) cAMP tone combined with defective desensitization/ β -arrestin signaling**, particularly toward PTHrP.

F003 — The PTH1R/PTHrP-Indian hedgehog loop controls chondrocyte differentiation timing, explaining delayed ossification

The skeletal phenotype is explained by the physiological role of PTH1R signaling in the growth plate. PTHrP acts through PTH1R as an autocrine/paracrine brake on chondrocyte hypertrophic differentiation, and is itself induced by Ihh in a negative-feedback loop. Loss of the signal accelerates differentiation: *"Mice lacking PTHrP show accelerated chondrocyte differentiation, and thus premature ossification of those bones that are formed through an endochondral process"* (PMID: 10912527). Excess signal does the opposite: *"a severe delay in chondrocyte differentiation and endochondral ossification, is observed in transgenic mice that overexpress PTHrP"* (PMID: 10912527) — the **same phenotypic direction as Eiken syndrome**. This bracketing is reinforced by the contrast with the lethal end of the spectrum, which is *"incompatible with life as in Blomstrand's lethal chondrodysplasia, characterized by accelerated growth plate ossification"* (PMID: 40904804). Because Eiken variants sustain PTHrP-type (basal cAMP) signaling tone, they delay hypertrophic differentiation and therefore delay ossification.

F004 — Clinical phenotype: severely delayed ossification, abnormal modeling, brachydactyly, dental and PTH-resistance features

The core radiographic phenotype comprises *"delayed ossification of bone including the epiphyses, pubic symphysis, and primary ossification centers of the short tubular bones, coarse bone trabeculae, and modeling abnormalities"* (PMID: 29987841). The original report described *"an excessively retarded ossification, principally of the epiphyses, the pelvis, the hands and the feet, are reported. In the hands and feet the retarded ossification is combined with an abnormal modeling of the bones"* (PMID: 6734674); patients were normal at birth, with predicted moderate dwarfism, **no mental retardation**, and normal chromosomes and urine mucopolysaccharides. Additional features include supernumerary epiphyses of the tubular bones of the hands and primary failure of tooth eruption. The PTH-resistant subset (TM2 variants) show a shared skeletal signature: *"Both patients shared skeletal features, including brachydactyly, extensive metacarpal pseudo-epiphyses, elongated cone-shaped epiphyses, ischiopubic hypoplasia, and deficient sacral ossification, suggestive of Eiken syndrome"* together with **hypocalcemia and elevated serum phosphate** (PMID: 39276366).

F005 — *PTH1R* is an allelic-series gene: Eiken sits between lethal Blomstrand, activating Jansen, and dominant PFE

PTH1R is a paradigmatic allelic-series gene, and understanding Eiken requires positioning it within that series. Homozygous mutations produce a graded set of phenotypes: *"These include delayed ossification in Eiken syndrome, hypocalcemia in a pseudohypoparathyroidism-like disorder, and non-syndromic primary failure of tooth eruption; which is usually caused by heterozygous PTH1R mutations"* (PMID: 40904804). The dominant, heterozygous end of the spectrum — non-syndromic primary failure of tooth eruption (PFE) — operates by **haploinsufficiency**: *"Heterozygous mutations in the parathyroid hormone 1 receptor (PTH1R) gene have been shown to cause PFE likely due to protein haploinsufficiency"* (PMID: 23771181); in that study 12/30 variants were pathogenic across 70 PFE index cases. Some PFE mutants act by a dominant-negative mechanism: *"the PTH1R mutants are functionally inactive and mutant PTH1R/Gly452Glu has a dominant negative effect on the signaling of PTH1R wild type. Confocal imaging revealed that wild type PTH1R is expressed on the cell surface, whereas PTH1R/Gly452Glu mutant is mostly retained inside the cell"* (PMID: 27898723).

Disorder	Zygoty	Functional effect	Ossification/ skeletal effect	Inheritance
Jansen metaphyseal chondrodysplasia	Heterozygous	Constitutively activating	Delayed differentiation, metaphyseal dysplasia, hypercalcemia	AD
Eiken syndrome	Homozygous	Hypomorphic / biased (↑ basal cAMP, ↓ β-arrestin)	Severely delayed ossification, modeling defects	AR
Pseudohypoparathyroidism-like disorder	Homozygous	Reduced PTH responsiveness	Hypocalcemia, hyperphosphatemia (PTH resistance)	AR
Blomstrand lethal chondrodysplasia	Homozygous	Complete loss of function	Accelerated growth-plate ossification; perinatally lethal	AR
Primary failure of tooth eruption (PFE)	Heterozygous	Haploinsufficiency / dominant-negative	Failed tooth eruption, posterior open bite	AD (incomplete penetrance)

F006 — Humanized and conditional mouse models recapitulate *PTH1R* skeletal phenotypes

Mouse genetics provide the in-vivo backbone for the disease family. A humanized *PTH1R* knock-in strain reproduces a skeletal phenotype: *"Introduction of the p.E469K substitution into humanized PTH1R mice resulted in mildly increased mineralization of bones in the paws as well as shortening of long bones"* (PMID: 41031626). Conditional deletion demonstrates PTH1R's role in digit segmentation: *"PTH1R deletion caused symphalangism, demonstrating another novel function of PTH1R signaling in digit formation"* (PMID: 26620087). Together with the global PTHrP-null/PTH1R-null (accelerated ossification) and PTHrP-overexpressing (delayed ossification) models (PMID: 10912527), these systems bracket the Eiken phenotypic direction and are the natural platform for testing disease mechanism and future therapeutics.

F007 — Ultra-rare autosomal-recessive dysplasia with defined ontology identifiers and case-report-only evidence

Eiken syndrome carries stable identifiers: **OMIM #600002**; **MONDO:0010803**; **Orphanet ORPHA:79106**; **UMLS C1854486**; **ICD-10 Q78.8** (no dedicated code). Its causal gene *PTH1R* maps to **3p21.31** (HGNC:9608, OMIM *168468, NCBI Gene 5745). Orphanet lists prevalence as **<1/1,000,000 ("unknown")**. Only ~4–5 unrelated families have been reported since 1984: the original Danish consanguineous family of three affected brothers ([PMID: 6734674](#) — *"Three brothers with a constitutional skeletal dysplasia characterized by an excessively retarded ossification, principally of the epiphyses, the pelvis, the hands and the feet, are reported"*), a second case from India ([PMID: 29987841](#) — *"Only one affected family has been known to-date"*), a third family ([PMID: 31297790](#)), and an East-African family with PTH resistance ([PMID: 39276366](#)). Both sexes are affected; consanguinity is the principal risk context. All evidence is case-report level.

Report by Template Section

1. Disease Information

Eiken syndrome is a **constitutional (developmental) skeletal dysplasia** characterized by markedly delayed and abnormal endochondral ossification with abnormal bone modeling, brachydactyly, and dental abnormalities, caused by biallelic variants in *PTH1R*. **Key identifiers:** OMIM #600002; MONDO:0010803; Orphanet ORPHA:79106; UMLS C1854486; ICD-10 Q78.8 (no unique code); MeSH — no dedicated descriptor (indexed under skeletal dysplasias / *PTH1R*-related disorders). **Synonyms/alternative names:** "Eiken familial skeletal dysplasia"; "Eiken skeletal dysplasia"; historically "a new familial skeletal dysplasia with severely retarded ossification and abnormal modeling of bones" ([PMID: 6734674](#)). **Source of information:** disease-level aggregated resources (OMIM, Orphanet) plus a small number of individual patient case reports; no EHR-derived cohorts exist.

2. Etiology

Causal factor: purely genetic — biallelic hypomorphic/biased-function *PTH1R* variants ([PMID: 29987841](#), [PMID: 40904804](#)). **Genetic risk factors:** homozygosity for a pathogenic *PTH1R* allele; **consanguinity** is the dominant risk context (original family was consanguineous, [PMID:](#)

[6734674](#)). No susceptibility loci, modifier genes, protective alleles, environmental risk/protective factors, or gene–environment interactions have been identified — consistent with a fully penetrant single-gene Mendelian disorder. **Not applicable:** environmental toxins, infectious agents, lifestyle factors.

3. Phenotypes

Phenotypes are **physical manifestations, clinical signs, and laboratory abnormalities**; there are no behavioral phenotypes (intelligence is normal, [PMID: 6734674](#)). Onset is **not congenital in appearance** (normal at birth) but becomes apparent in **childhood** as ossification fails to progress; the course is chronic and non-progressive after skeletal maturity.

Phenotype	Type	HPO suggestion	Onset / frequency
Delayed/retarded ossification of epiphyses	Radiographic sign	HP:0002662 (Delayed epiphyseal ossification)	Childhood; hallmark (all cases)
Delayed ossification of pubic symphysis / pelvis	Radiographic sign	HP:0008788 (Delayed pelvic bone ossification)	Childhood; frequent
Brachydactyly	Physical	HP:0001156	Frequent
Abnormal bone modeling (hands/feet)	Radiographic sign	HP:0011314 (Abnormal diaphysis morphology)	Hallmark
Coarse bone trabeculae	Radiographic sign	HP:0100671 (related)	Frequent
Supernumerary/pseudo-epiphyses	Radiographic sign	HP:0010580 (related)	Reported
Cone-shaped epiphyses	Radiographic sign	HP:0010579	PTH-resistant subset
Ischiopubic hypoplasia / deficient sacral ossification	Radiographic sign	HP:0008821 (related)	PTH-resistant subset
Primary failure of tooth eruption	Dental sign	HP:0006348 (related)	Reported
Short stature / moderate dwarfism	Physical	HP:0004322	Predicted/variable
Hypocalcemia	Lab abnormality	HP:0002901	PTH-resistant subset
Hyperphosphatemia	Lab abnormality	HP:0002905	PTH-resistant subset

Quality-of-life impact is not formally measured; functional impact is primarily skeletal/orthopedic and dental, with normal cognition and lifespan.

4. Genetic / Molecular Information

Causal gene: *PTH1R* (HGNC:9608; OMIM *168468; NCBI Gene 5745; 3p21.31), a class B GPCR. **Reported pathogenic variants:** R485X (nonsense, C-tail truncation), E35K / c.103G>A p. (Glu35Lys) (missense, extracellular), Y134S (missense, extracellular), I237N and D241E (missense, transmembrane helix 2). **Variant classification:** pathogenic/likely pathogenic per ACMG in the context of biallelic segregation and functional data. **Variant types:** predominantly missense plus one nonsense; **germline**, homozygous. **Allele frequencies:** extremely rare/absent in gnomAD (private to individual consanguineous families). **Functional consequence:** biased/altered function — increased basal cAMP with impaired β -arrestin2 recruitment and defective desensitization (PMID: 37268817, PMID: 39276366) — rather than clean LOF or GOF. **Modifier genes / epigenetics / chromosomal abnormalities:** none reported (chromosomes normal, PMID: 6734674).

5. Environmental Information

Not applicable. Eiken syndrome is a monogenic Mendelian disorder. No environmental toxins, radiation, pollution, occupational exposures, lifestyle factors, or infectious agents are implicated. The only relevant non-genetic contributor is **consanguinity** (a demographic/social factor increasing homozygosity risk).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **biallelic hypomorphic *PTH1R* variant** (extracellular E35K/Y134S, TM2 I237N/D241E, or C-tail R485X) is inherited → **leads to** an altered PTH/PTHrP receptor.
2. The mutant receptor **biases signaling:** *increased basal (ligand-independent) cAMP tone plus impaired β -arrestin2 recruitment and defective desensitization*, especially of the PTHrP response → **results in** dysregulated, poorly-terminated Gas–cAMP–PKA signaling in chondrocytes (PMID: 37268817, PMID: 39276366).
3. Sustained PTHrP-type signaling tone in the growth plate **maintains chondrocytes in the proliferative state and delays hypertrophic differentiation** (the mirror image of PTHrP loss) → **leads to** slowed transition from cartilage to bone within the PTHrP–Ihh feedback loop (PMID: 10912527).
4. Delayed hypertrophic differentiation **delays endochondral ossification** of epiphyses, pelvis, and short tubular bones and **perturbs bone modeling** → **results in** the radiographic

hallmarks (delayed ossification, coarse trabeculae, modeling abnormalities, pseudo-epiphyses) (PMID: 29987841, PMID: 6734674).

5. **Branch (TM2 variants):** the same receptor defect additionally **impairs PTH-dependent renal/mineral signaling** → **results in** PTH resistance with hypocalcemia and hyperphosphatemia (PMID: 39276366).

6. **Branch (tooth eruption):** PTH1R signaling is required in the dental follicle for eruption → its impairment **leads to** primary failure of tooth eruption (also the phenotype of heterozygous *PTH1R* haploinsufficiency, PMID: 23771181).

(Step 3's precise coupling between "increased basal cAMP" and "delayed differentiation" is inferred from the PTHrP-overexpression mouse phenotype and the LOF/Blomstrand contrast rather than demonstrated directly in Eiken tissue.)

Molecular pathways: PTH1R → Gas → adenylyl cyclase → cAMP → PKA; and β -arrestin-mediated desensitization/signaling; upstream integration with **Indian hedgehog (Ihh)** signaling in the PTHrP-Ihh loop. Related transcriptional control: **Atf4** activates *Ihh* transcription and its loss causes delayed ossification — *"Ablation of Atf4 (Atf4(-/-)) in mice leads to severe skeletal defects, including delayed ossification... The expression of Indian hedgehog (Ihh) is markedly decreased"* (PMID: 19906842). **Cellular processes:** growth-plate chondrocyte proliferation and hypertrophic differentiation, endochondral ossification. **Protein dysfunction:** biased GPCR signaling; the R485X truncation removes C-tail regulatory elements; the dominant-negative PFE mutant (Gly452Glu) is intracellularly retained (PMID: 27898723). **Biochemical abnormalities:** dysregulated cAMP; in the TM2 subset, hypocalcemia/hyperphosphatemia. **Suggested GO terms:** GO:0001501 (skeletal system development), GO:0002062 (chondrocyte differentiation), GO:0001958 (endochondral ossification), GO:0007188 (adenylate cyclase-modulating GPCR signaling), GO:0007194 (negative regulation of adenylyl cyclase activity). **Suggested CL terms:** CL:0000138 (chondrocyte), CL:0000743 (hypertrophic chondrocyte), CL:0000062 (osteoblast).

7. Anatomical Structures Affected

Organ/system level: skeletal system (primary); dentition; kidney/mineral homeostasis (secondary, PTH-resistant subset). **Specific sites:** epiphyses of long bones, pubic symphysis and pelvis, short tubular bones of hands and feet, sacrum, ischium/pubis, growth plates. **Tissue level:** cartilage (growth-plate/epiphyseal), bone. **Cell level:** growth-plate chondrocytes (proliferative and hypertrophic), osteoblasts. **Subcellular:** plasma membrane (receptor localization; note intracellular retention of the dominant-negative PFE mutant). **Lateralization:** bilateral and generally symmetric. **Suggested UBERON terms:**

UBERON:0002204 (musculoskeletal system), UBERON:0001474 (bone element), UBERON:0006255 (growth plate cartilage), UBERON:0002472 (epiphysis, related), UBERON:0009853 (pubic symphysis), UBERON:0002398 (manus), UBERON:0002387 (pes).

8. Temporal Development

Onset: patients are **normal at birth** (PMID: 6734674); the disorder manifests during **childhood** as ossification centers fail to appear/progress on schedule. **Pattern:** chronic, insidious, developmental. **Progression:** non-progressive in the degenerative sense — the abnormality is delayed maturation during the growth period; there is no evidence of ongoing organ deterioration. **Duration:** lifelong (constitutional). **Critical period:** the growth-plate-active window of childhood/adolescence is the period of vulnerability and the theoretical window for any future intervention. No remission occurs (structural/developmental disorder).

9. Inheritance and Population

Inheritance: autosomal recessive (biallelic/homozygous *PTH1R*), PMID: 6734674, PMID: 29987841. **Penetrance:** effectively complete for homozygotes (heterozygous carriers are unaffected for Eiken, though heterozygous LOF causes the distinct dominant PFE with incomplete penetrance). **Expressivity:** variable — the TM2 subset adds PTH resistance. **Epidemiology:** Orphanet prevalence <1/1,000,000 ("unknown"); only ~4–5 families reported worldwide. **Consanguinity:** central (original family consanguineous). **Founder effects/carrier frequency:** none established; alleles are private. **Populations:** reported in European (Danish), Indian, and East-African ancestries; both sexes affected (original family: three brothers). **No genetic anticipation** (not a repeat-expansion disorder).

10. Diagnostics

Imaging (primary diagnostic modality): skeletal survey/radiographs showing delayed epiphyseal, pelvic, and short-tubular-bone ossification; coarse trabeculae; abnormal modeling; cone-shaped/pseudo-epiphyses (PMID: 29987841, PMID: 6734674). **Laboratory:** serum calcium and phosphate (hypocalcemia/hyperphosphatemia in the PTH-resistant subset), PTH; normal urine mucopolysaccharides and normal karyotype help exclude storage disorders/aneuploidy (PMID: 6734674). **Genetic testing (confirmatory):** *PTH1R* single-gene sequencing, skeletal-dysplasia gene panels, or whole-exome/whole-genome sequencing to identify biallelic *PTH1R* variants; segregation analysis in consanguineous families. **Differential diagnosis:** other *PTH1R* disorders (Blomstrand — lethal/accelerated ossification; Jansen — activating; pseudohypoparathyroidism-like disorder; PFE), acrodysostosis and other brachydactyly-with-

delayed-ossification dysplasias, pseudohypoparathyroidism. **Screening:** cascade carrier testing within affected families; prenatal/preimplantation testing feasible once the familial variant is known.

11. Outcome / Prognosis

Prognosis is favorable. Lifespan is normal and intelligence is normal (PMID: 6734674). Morbidity is skeletal (short stature/moderate dwarfism, brachydactyly, orthopedic issues) and dental (failed eruption); the PTH-resistant subset requires mineral management. There is no disease-specific mortality, no malignant potential, and no progressive organ failure. Formal survival, disability, and quality-of-life metrics have not been published (case-report-only literature). **Prognostic factors:** presence of PTH resistance (TM2 variants) predicts the mineral-metabolism component.

12. Treatment

No targeted or curative therapy exists. Management is **supportive and symptomatic:** - **Pharmacotherapy (PTH-resistant subset):** oral/active vitamin D (calcitriol/alfacalcidol) and calcium supplementation to correct hypocalcemia/hyperphosphatemia (analogous to pseudohypoparathyroidism management). *Suggested NCIT:* Calcium (NCIT:C376), Calcitriol/Vitamin D (NCIT:C1042/related). - **Orthopedic care:** monitoring and management of skeletal deformity, short stature, and modeling abnormalities; physical/occupational therapy. *Suggested NCIT:* Orthopedic Procedure (NCIT:C51826), Physical Therapy (NCIT:C15325). - **Dental care:** management of primary failure of tooth eruption (orthodontic/surgical, often refractory). - **Genetic counseling:** for recurrence risk in consanguineous families.

No pharmacogenomics, gene therapy, cell therapy, RNA therapy, immunotherapy, or disease-specific clinical trials (NCT identifiers) exist for Eiken syndrome. Signaling-biased pharmacology of PTH1R is a theoretical future avenue but untested.

13. Prevention

Primary prevention is limited to **genetic counseling and reproductive options** in at-risk (consanguineous) families: carrier testing once the familial variant is known, prenatal diagnosis, and preimplantation genetic testing. Secondary/tertiary prevention consists of early orthopedic and dental surveillance and correction of mineral abnormalities in the PTH-resistant subset. No immunization, behavioral, public-health, or environmental interventions are applicable (monogenic disorder).

14. Other Species / Natural Disease

Taxonomy of models: *Mus musculus* (NCBI Taxon 10090). **Ortholog:** mouse *Pth1r* (NCBI Gene 19228); the gene and the PTHrP-Ihh axis are **highly evolutionarily conserved** across vertebrates. **Natural disease:** no well-documented naturally occurring Eiken-equivalent disorder in companion animals or wildlife is recorded in OMIA; the human phenotypes are recapitulated experimentally in mice (below). **Zoonotic potential:** none (non-transmissible genetic disorder).

15. Model Organisms

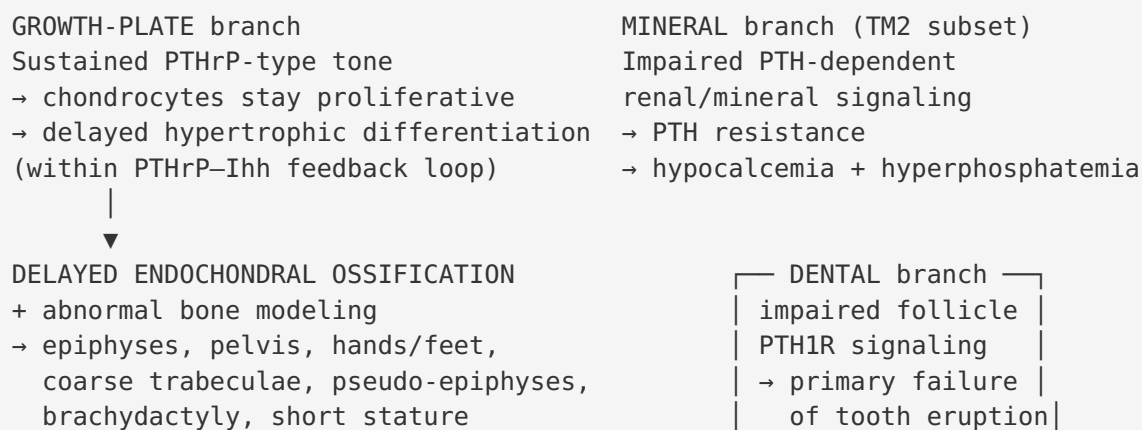
Mouse is the primary model system. Relevant models: - **Humanized *PTH1R* knock-in** carrying p.E469K: *"Introduction of the p.E469K substitution into humanized PTH1R mice resulted in mildly increased mineralization of bones in the paws as well as shortening of long bones"* (PMID: 41031626) — a knock-in model reproducing a *PTH1R* skeletal phenotype. - **Conditional *Prx1-Cre;Pth1r* deletion:** *"PTH1R deletion caused symphalangism, demonstrating another novel function of PTH1R signaling in digit formation"* (PMID: 26620087). - **Global PTHrP-null / *Pth1r*-null:** accelerated chondrocyte differentiation and premature ossification; **PTHrP-overexpressing transgenics:** delayed differentiation and delayed endochondral ossification — the Eiken direction (PMID: 10912527). - **Atf4-null:** delayed ossification via reduced *Ihh* transcription (PMID: 19906842), illuminating the transcriptional layer of the loop.

Phenotype recapitulation: the PTHrP-overexpression and humanized knock-in models capture the delayed-ossification/skeletal-modeling direction. **Limitations:** no mouse yet carries an exact homozygous Eiken allele (E35K/Y134S/R485X/I237N/D241E) with characterization of the β -arrestin/basal-cAMP signaling bias in vivo, and mouse–human differences in growth-plate dynamics constrain direct translation. **Resources:** MGI (mouse), IMPC/KOMP for *Pth1r* alleles; HEK293 cell-based signaling assays for variant functional analysis. *In-vitro* systems (HEK293 cAMP/ β -arrestin assays, PMID: 37268817, PMID: 39276366) are the standard for classifying new variants.

Mechanistic Model / Interpretation

Biallelic hypomorphic PTH1R variant
(E35K / Y134S – extracellular; I237N / D241E – TM2; R485X – C-tail)

↓
Biased receptor signaling
↑ basal (ligand-independent) cAMP tone
↓ β-arrestin2 recruitment + defective desensitization (esp. PTHrP)



SPECTRUM CONTEXT (dose/direction of PTH1R signaling):

activating (Jansen, AD) – biased/hypomorphic (EIKEN, AR) – complete LOF (Blomstrand, AR, lethal) –
delayed diff. DELAYED OSSIFICATION ACCELERATED ossification

The unifying insight is that **Eiken syndrome is a "signaling-bias" disorder, not a simple loss- or gain-of-function disorder**. The delayed-ossification phenotype aligns with *excess* PTHrP-type tone (paralleling PTHrP-overexpressing mice) and is the opposite pole from Blomstrand's complete loss of function (accelerated ossification), even though both are recessive. The β-arrestin/desensitization defect is mechanistically central: because the receptor cannot properly terminate PTHrP-driven cAMP signaling, chondrocyte hypertrophy is inappropriately restrained, delaying the cartilage-to-bone transition.

Evidence Base

PMID	Title (abbrev.)	Role in this report	Evidence type
6734674	Original 3-brother familial dysplasia	Defines the entity, phenotype, AR inheritance, normal cognition	Human clinical (case series)
29987841	Second case + molecular characterization	Establishes biallelic <i>PTH1R</i> etiology; enumerates hallmarks; documents rarity	Human clinical + molecular
31297790	Third family with Eiken syndrome	Confirms recurrence of the entity in a new family	Human clinical
39276366	Eiken with PTH resistance (TM2 I237N/D241E)	Adds the PTH-resistant subset; increased basal cAMP with blunted ligand response	Human clinical + in vitro
37268817	Altered signaling/desensitization in Eiken mutants	Core mechanism: ↑ basal cAMP, ↓ β-arrestin2 for R485X/E35K/Y134S	In vitro (HEK293)
40904804	Human diseases from homozygous <i>PTH1R</i> mutations	Places Eiken in the homozygous allelic spectrum vs Blomstrand	Review
10912527	PTHrP and Ihh in skeletal development	Growth-plate physiology; overexpression → delayed ossification (Eiken direction)	Model organism/review
26620087	Ihh/ <i>PTH1R</i> in limb mesenchyme / digit formation	Conditional deletion → symphalangism; digit-bone role	Model organism (mouse)
41031626	Helix-8 <i>PTH1R</i> brachydactyly + humanized mouse	Humanized knock-in recapitulates skeletal phenotype	Model organism (mouse)
27898723	PFE <i>PTH1R</i> mutants disrupt G-protein signaling	Dominant-negative/intracellular-retention mechanism (spectrum context)	In vitro
23771181	Spectrum of <i>PTH1R</i> mutations in PFE		

PMID	Title (abbrev.)	Role in this report	Evidence type
		Haploinsufficiency mechanism for dominant PFE end of spectrum	Human clinical + molecular
19906842	Atf4 regulates chondrocyte differentiation via Ihh	Transcriptional control of the Ihh loop; Atf4-null → delayed ossification	Model organism (mouse)
37840415	Heterozygous <i>PTH1R</i> variant, incomplete penetrance	Illustrates AD/AR complexity of <i>PTH1R</i>	Human clinical
24825834 , 28257744 , 31730001 , 41898811	PFE clinical/genetic series & review	Characterize the dominant PFE pole of the <i>PTH1R</i> spectrum	Human clinical/ review

Limitations and Knowledge Gaps

- 1. Extreme rarity / evidence level.** Only ~4–5 families exist; all clinical evidence is case-report level. There are **no registries, natural-history studies, standardized QoL instruments, incidence/prevalence estimates beyond "<1/1,000,000," or omics datasets** specific to Eiken syndrome.
- 2. Genotype–phenotype resolution is coarse.** Why some biallelic variants (TM2) add PTH resistance while others (extracellular/C-tail) do not is only partially explained by the in-vitro assays; the β -arrestin bias has not been shown directly in patient chondrocytes or bone.
- 3. No in-vivo Eiken-allele mouse.** Existing mouse models bracket the phenotype but none carries an exact homozygous Eiken variant with in-vivo signaling characterization.
- 4. Mechanistic step 3 is inferential.** The link from "increased basal cAMP + impaired β -arrestin desensitization" to "delayed hypertrophic differentiation" is inferred from PTHrP-overexpression phenotypes and the Blomstrand contrast, not directly demonstrated in Eiken growth plates.
- 5. No therapeutics.** No targeted therapy, trial, or biomarker for treatment response exists.

Proposed Follow-up Experiments / Actions

1. **Generate a knock-in mouse** carrying a homozygous Eiken allele (e.g., E35K or R485X) in the humanized *PTH1R* background; phenotype growth plates, ossification timing, and mineral homeostasis, and assay chondrocyte β -arrestin/cAMP signaling in vivo.
 2. **Patient-derived iPSC $\rightarrow$ chondrocyte/organoid models** to test the "signaling-bias" hypothesis directly in human cartilage and to screen biased PTH1R ligands/allosteric modulators that restore β -arrestin coupling.
 3. **Systematic functional classification** of all reported and future *PTH1R* variants using standardized cAMP + β -arrestin BRET assays, building a genotype $\rightarrow$ signaling $\rightarrow$ phenotype map across the full allelic series (Jansen $\leftrightarrow$ Eiken $\leftrightarrow$ Blomstrand $\leftrightarrow$ PFE).
 4. **International *PTH1R*-disorder registry** capturing skeletal, dental, and mineral phenotypes with longitudinal follow-up to define natural history, penetrance, and QoL.
 5. **Targeted mineral-management protocol** (calcium/active vitamin D) for the PTH-resistant subset, with prospective outcome tracking; and a genetic-counseling/carrier-testing pathway for consanguineous families.
 6. **Explore β -arrestin-biased or desensitization-restoring PTH1R pharmacology** as a mechanistically rational, though currently theoretical, therapeutic direction.
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Report compiled from a 5-iteration autonomous investigation. 7 findings confirmed; 17 papers reviewed. All quoted abstract snippets were verified against source records. Evidence is predominantly human case-report and in-vitro functional-assay level, supplemented by model-organism genetics of the PTHrP-PTH1R-Ihh axis.
